



Estimating Vegetation Change from Fossil Pollen to Planetary Boundaries

A Global, Long-Term Synthesis Using Big Data



Scan the QR code to follow along!

 Ondřej Mottl

TIBS 2026





6/9 global Earth Planetary
Boundaries are already exceeded

(Richardson et al. 2023) *Science Advances*

Image generated by Bing



~20%

*Species to be exposed to **dangerous temperature** condition (with global increase of 2.0°C)*

IPCC AR6 Synthesis Report

Image by Justus Menke from Pexels



What if ...



... we inform the
conservation policies by
incorporating a **global,**
long-term synthesis on
ecosystem dynamics using
paleoecological **Big Data**?



 **Hi! I am Ondřej Mottl**



Assistant Professor at  Charles University, Prague, CZ

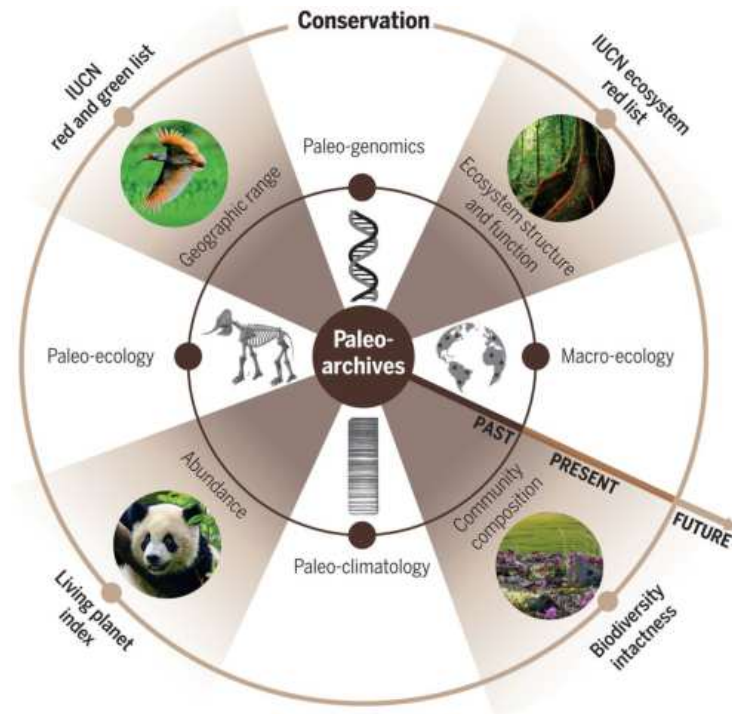
Interested in macroecology, palaeoecology, biodiversity, and data science

Head of the  Laboratory of Quantitative Ecology

-  [ondrej.mottl\(at\)natur.cuni.cz](mailto:ondrej.mottl(at)natur.cuni.cz)
-  ondrej.mottl.bsky.social
-  [Github](https://github.com/ondrej.mottl)
-  [ORCID](https://orcid.org/ondrej.mottl)
-  [Personal webpage](https://www.ondrej.mottl.cz)



Palaeoecology



Because **warning signals** ... are commonly identified by using time-series ... data, **paleo-archives** provide opportunities to **test conservation criteria** ...



Fossil pollen as a
window to the past
vegetation patterns

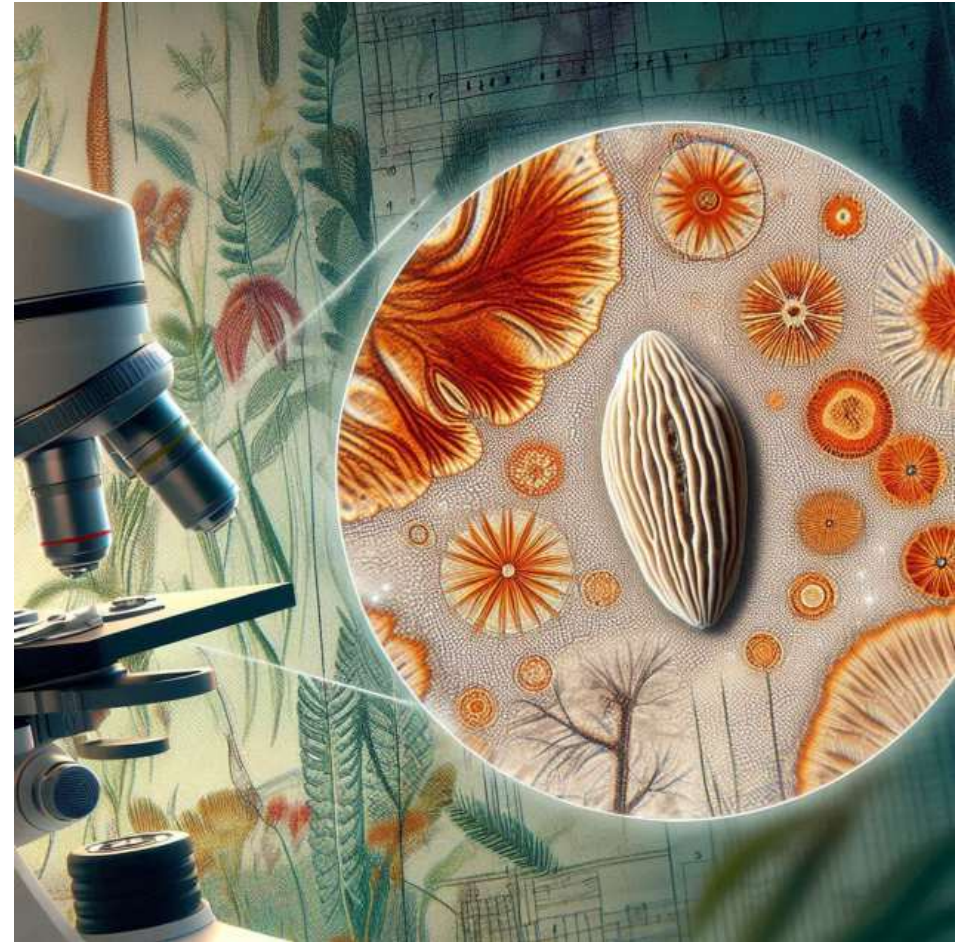


Image created by Bing



Neotoma Paleoecology Database

> 6650

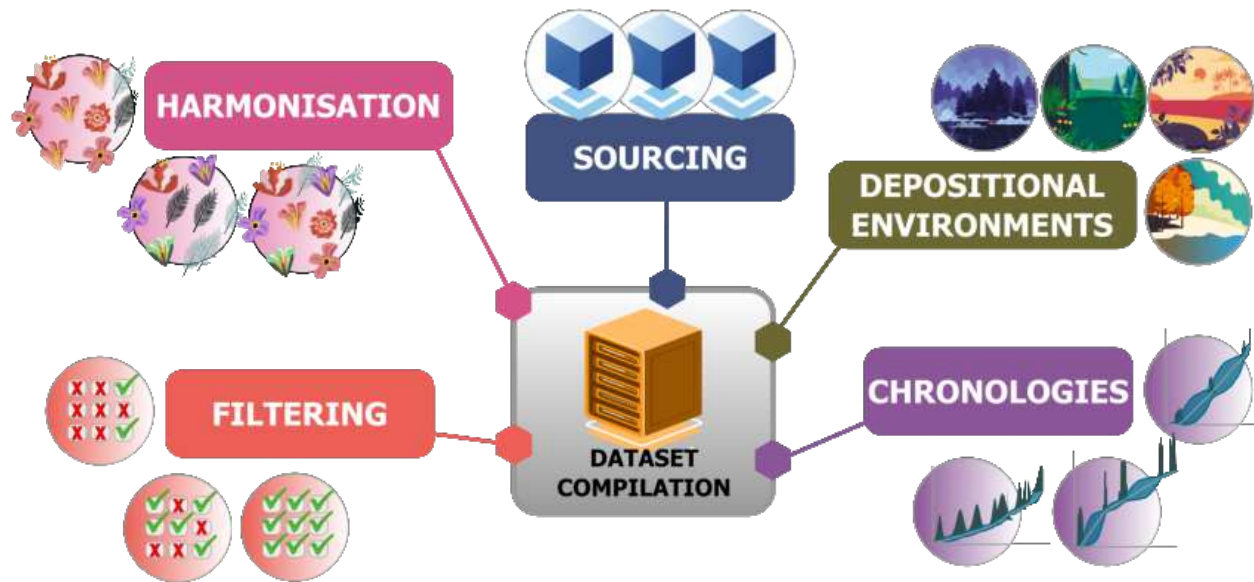
fossil pollen timeseries (records)



Fossil pollen datasets accessed from Neotoma on 2025-12-09



Palaeoecological dataset compilation



- Macro-scale data analysis
- Project specific
- Data quality control and reproducibility

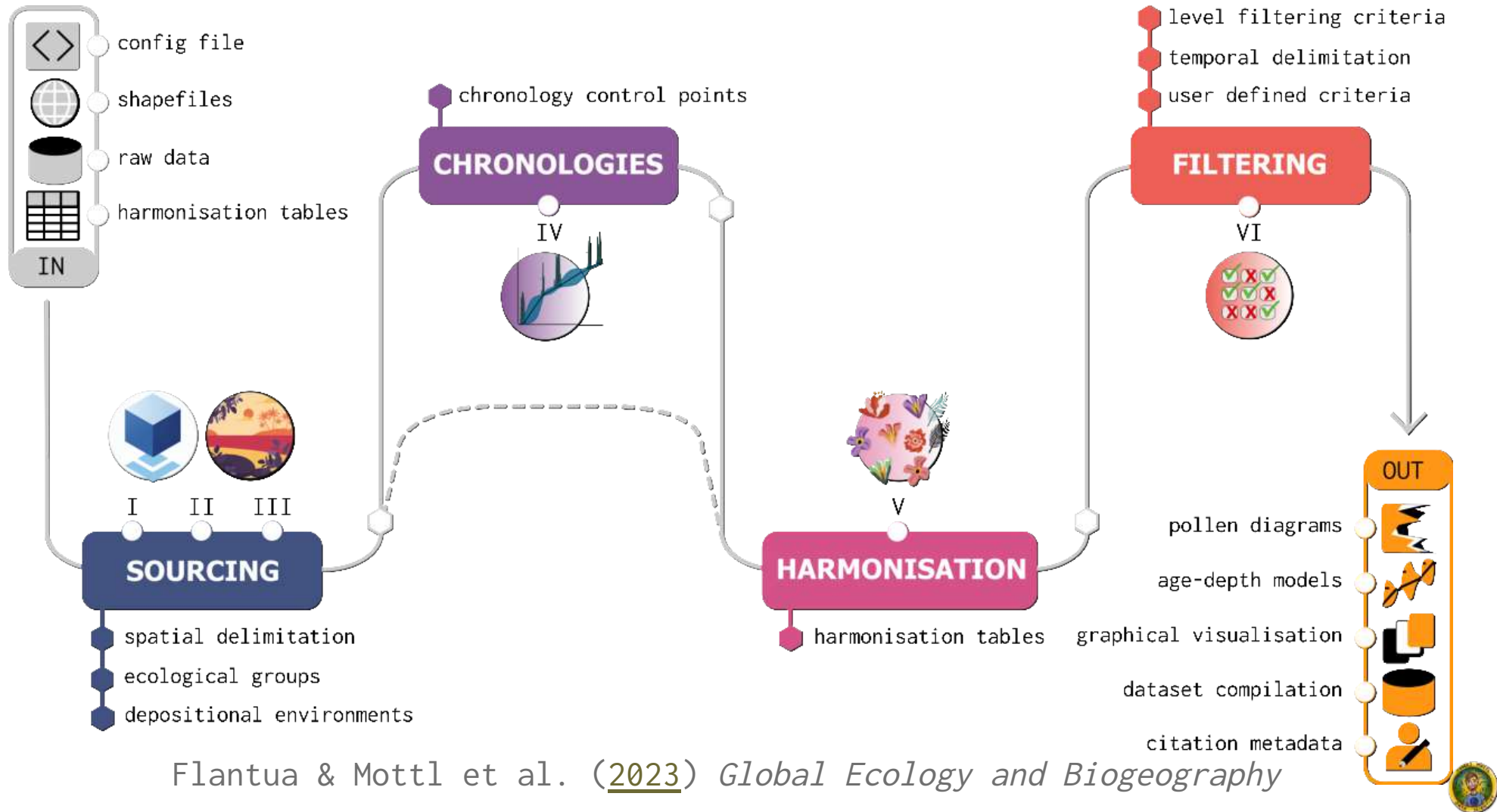




The workflow to process
global palaeoecological data
of fossil pollen for
vegetation-based
macroecological synthesis



FOSSILPOL workflow



Documentation (website)



PUBLISHED February 22, 2022
MODIFIED March 12, 2025

FOSSILPOL project

Summary

This website presents all information needed about the FOSSILPOL workflow that aims to process and standardise global palaeoecological pollen data.

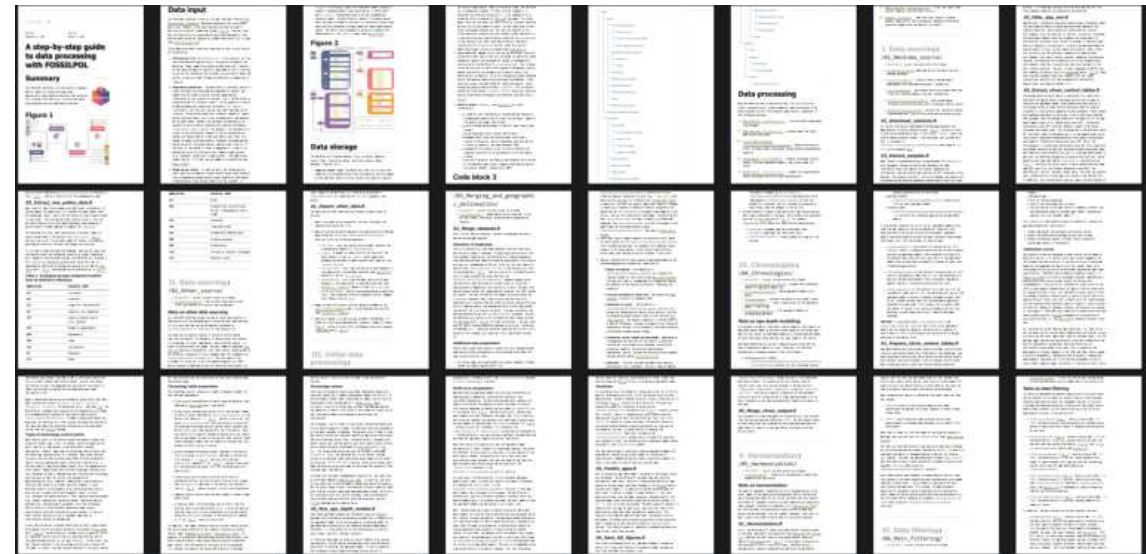
Authors

The FOSSILPOL workflow has been developed during the ERC project called the *Humans on Planet Earth (HOPE)* team at the *University of Bergen*:

Suzette G.A. Flantua*, Ondrej Mottl*, Vivian Felde*, Kuber P. Bhatta*, Hilary Birks, John-Arvid Grytnes, Alistair Seddon, H. John B. Birks

* shared first authors

[Ondrej Mottl](#) is the main maintainer of the FOSSILPOL project



Now bilingual! (English / Spanish)

Check it out! - bit.ly/FOSSILPOL



FOSSILPOL



- Open-source and reproducible workflow written in R
- Handles all the steps and flags places, where the user needs to specify their choices
- Modular structure to allow flexibility and customisation for different projects



**Global Ecology
and Biogeography**

A Journal of
Macroecology

METHOD Open Access

**A guide to the processing and standardization of global
palaeoecological data for large-scale syntheses using
fossil pollen**

[Suzette G. A. Flantua](#) [Ondrej Mottl](#) [Vivian A. Felde](#) [Kuber P. Bhatta](#) [Hilary H. Birks](#)
[John-Arvid Grytnes](#) [Alistair W. R. Seddon](#) [H. John B. Birks](#)

First published: 02 May 2023 | <https://doi.org/10.1111/geb.13693>



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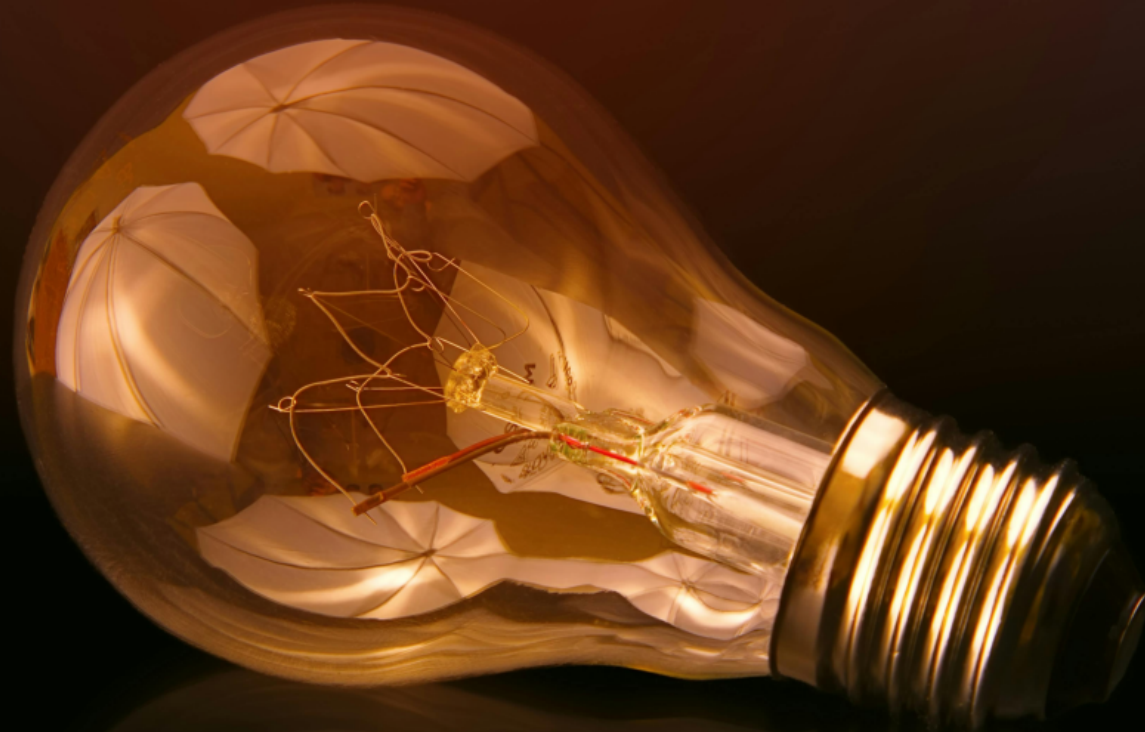
METHOD Open Access

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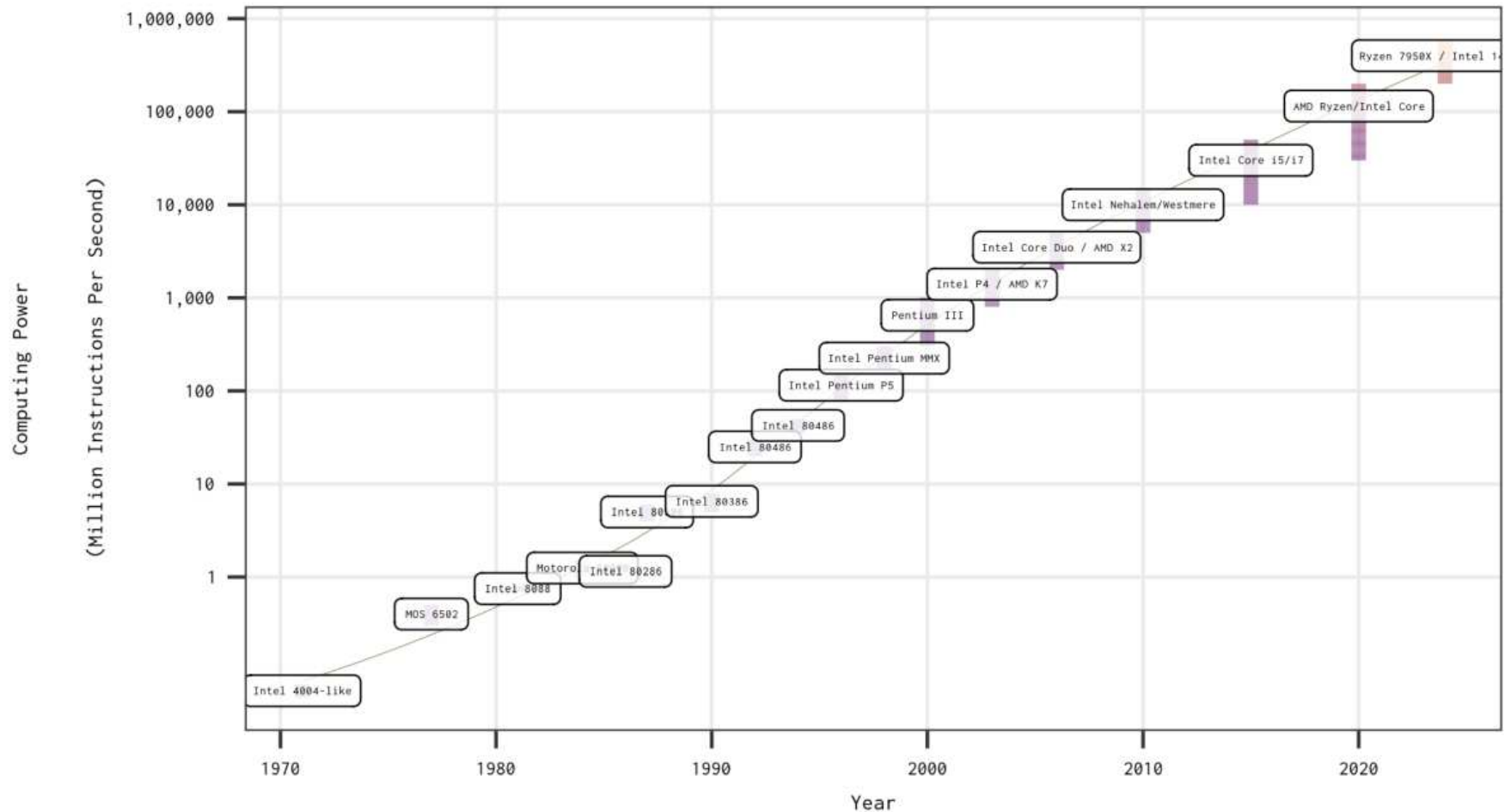


Inovations

Image by Johannes Plenio from Pexels



The power of a personal computer



The data obtained from [wikipedia/Instructions_per_second](https://en.wikipedia.org/wiki/Instructions_per_second)



Analytical palaeoecology



Advancements

Best practices and **reproducible workflows** for data standardization, Analyses that can accommodate **heterogeneous datasets**, Incentive structures that reward **software development**, Awareness of sampling context and biases, Best practices and conceptual frameworks to guide data integration, Development of taxon-free metrics, **Training** in quantitative methods and **Data Science**, Collaboration with data scientists, **Interdisciplinary** communication, collaboration, and funding



Analytical palaeoecology



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Best practices and reproducible workflows for data standardization, Analyses that can accommodate heterogeneous datasets, Incentive structures that reward software development, Awareness of sampling context and biases, Best practices and conceptual frameworks to guide data integration, Development of taxon-free metrics, Training in quantitative methods and Data Science, Collaboration with data scientists, Interdisciplinary communication, collaboration, and funding

Erin Dillon has talk directly after me!

Dillon et al. (2023) *Paleobiology*



Statistical advancements

New methods accounting for **effort**, **bias**, and **uncertainty**



Photo by [Andrea Piacquadio](#) from Pexels



Statistical advancements

New methods accounting for **effort**, **bias**, and **uncertainty**

- Hierarchical regression models
- Joined Species Distribution Models (jSDM)
- Model-based ordinations
- Copula methods
- ...



Photo by [Andrea Piacquadio](#) from Pexels



Statistical advancements

New methods accounting for **effort**, **bias**, and **uncertainty**

- Hierarchical regression models
- Joined Species Distribution Models (jSDM)
- Model-based ordinations
- Copula methods
- ...



More details in a moment in talk by **Gavin Simpson** in this Symposium!

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Estimating Rate of Change (RoC)

Issues with palaeoecological RoC
estimation

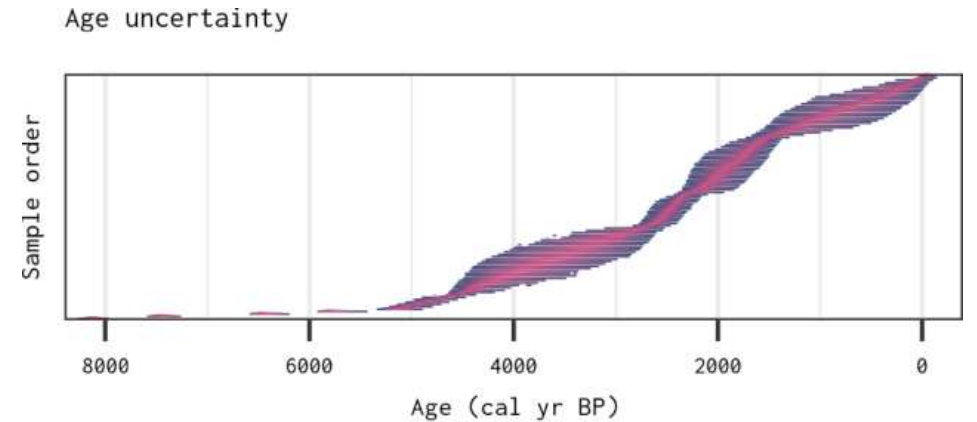


Estimating Rate of Change (RoC)

example data

Issues with palaeoecological RoC estimation

- Age uncertainties



Shading indicates the density of possible ages for each sample

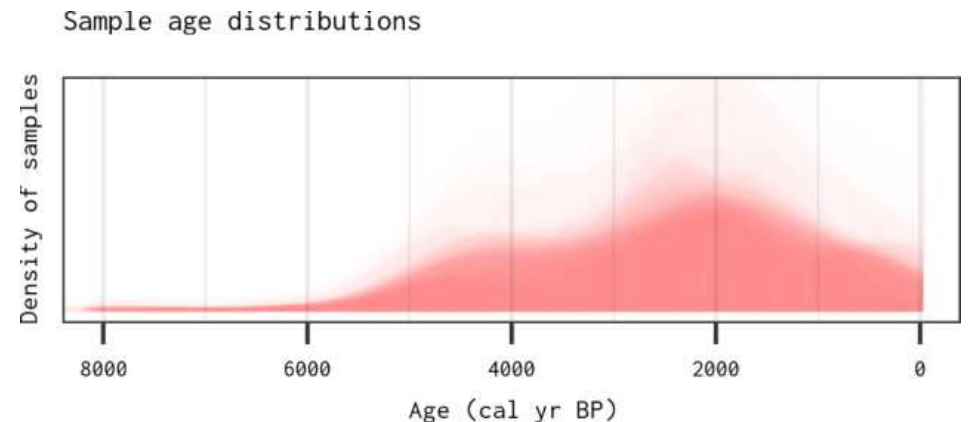
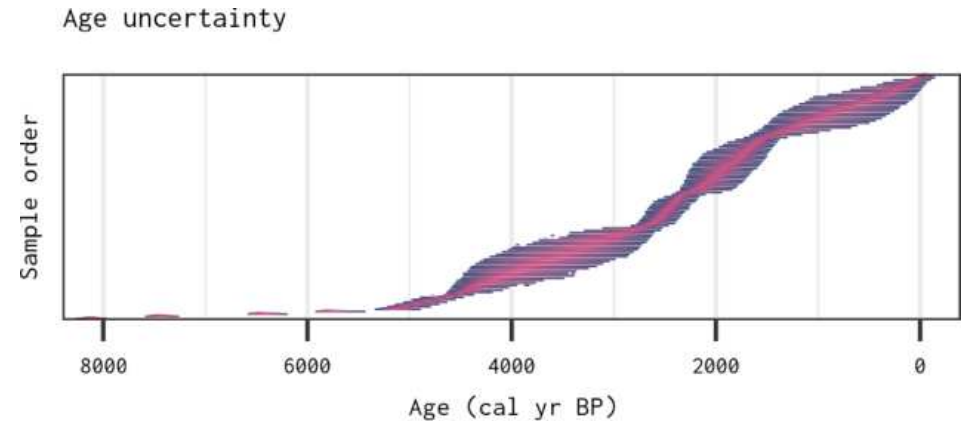


Estimating Rate of Change (RoC)

example data

Issues with palaeoecological RoC estimation

- Age uncertainties
- Uneven temporal distribution of samples

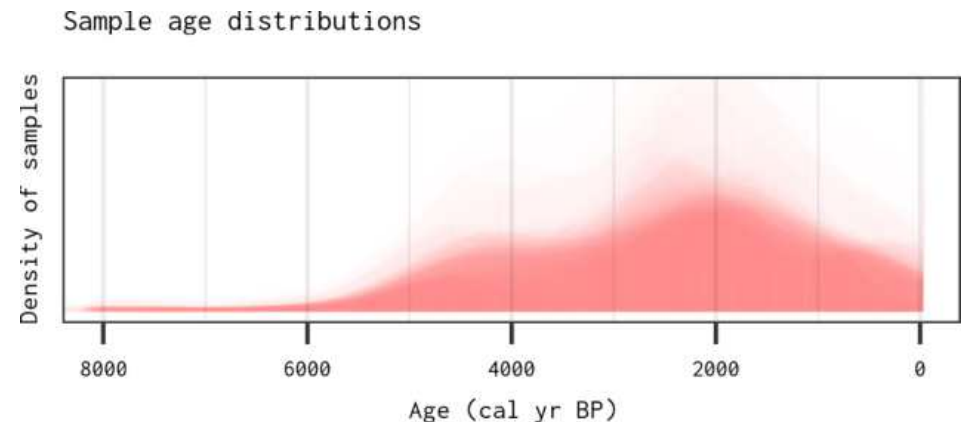
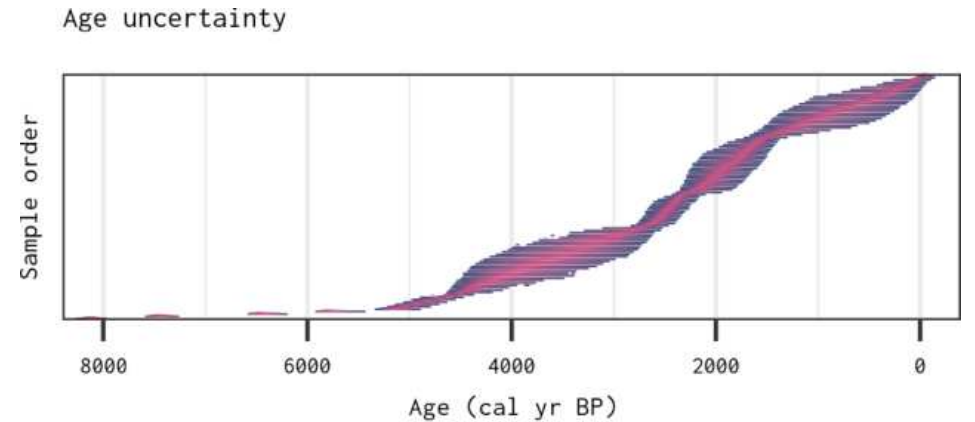


Estimating Rate of Change (RoC)

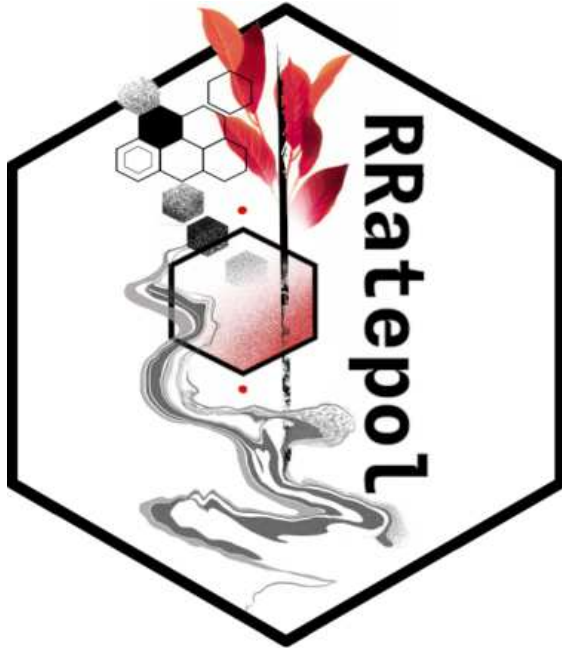
example data

Issues with palaeoecological RoC estimation

- Age uncertainties
- Uneven temporal distribution of samples
- Taxonomic uncertainties



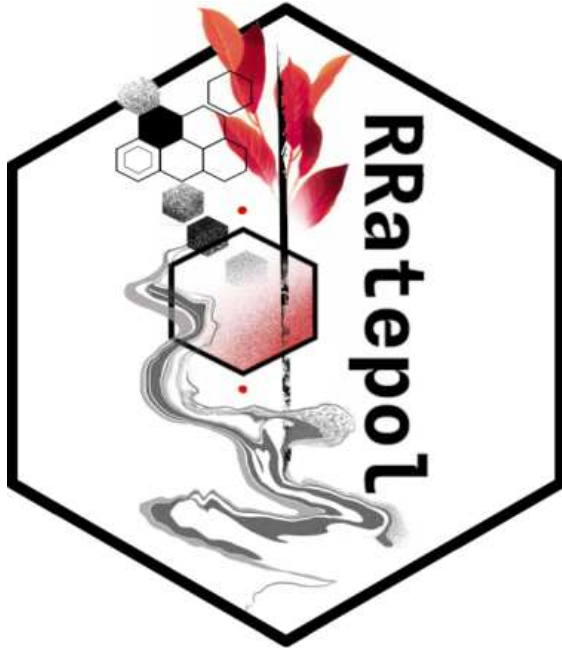
{RRatepol}



An R package for estimating **rate of change** (RoC) from **multivariate** (community) data in **time series**



{RRatepol}

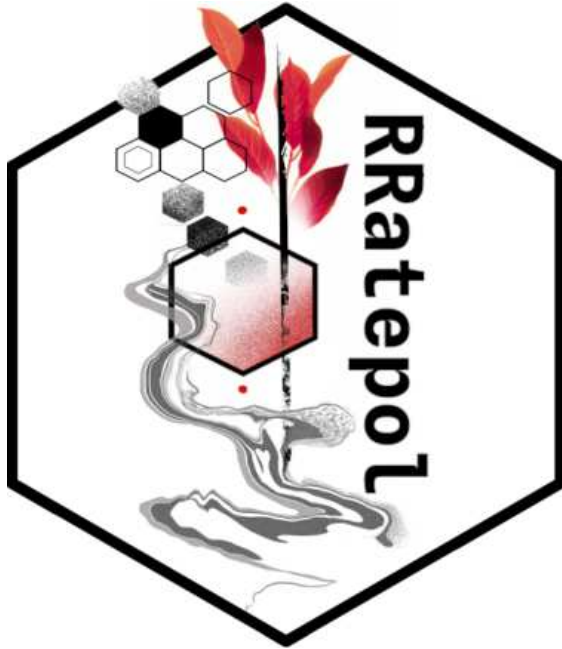


An R package for estimating **rate of change** (RoC) from **multivariate** (community) data in **time series**

- Handles **uneven time series**



{RRatepol}



An R package for estimating **rate of change** (RoC) from **multivariate** (community) data in **time series**

- Handles **uneven time series**
- **Propagates uncertainty** (age, taxonomic)



{RRatepol}



An R package for estimating **rate of change** (RoC) from **multivariate** (community) data in **time series**

- Handles **uneven time series**
- **Propagates uncertainty** (age, taxonomic)
- Can work with **various data types** (palaeoecological proxies)



{RRatepol}



Mottl et al. ([2021](#)) *Review of Palaeobotany and Palynology*



{RRatepol}



Mottl et al. ([2021](#)) *Review of Palaeobotany and Palynology*



{RRatepol}



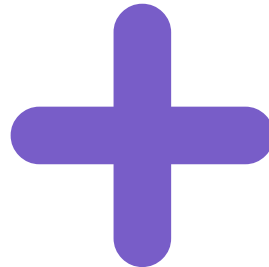
Mottl et al. ([2021](#)) *Review of Palaeobotany and Palynology*



Global rates of vegetation change over the past 18,000 years



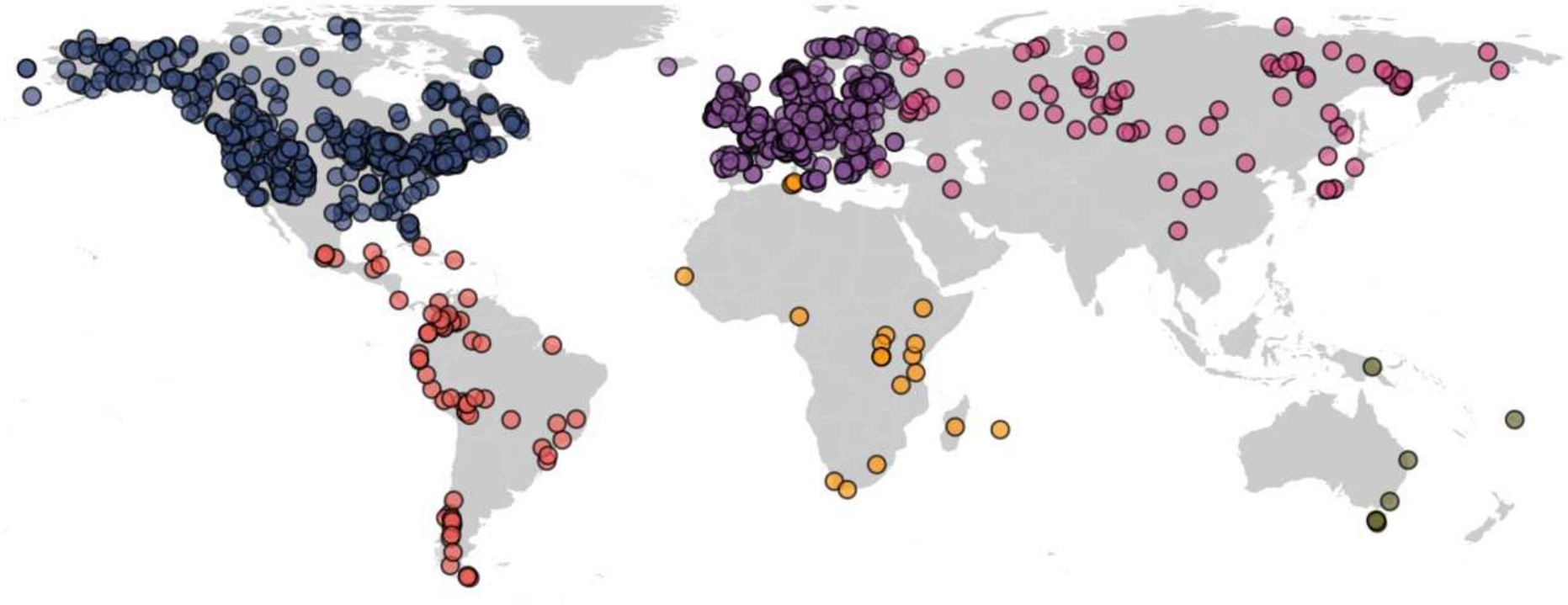
Global assebmly of fossil
pollen record



Estimation of rate of
vegetation change



Global rates of vegetation change



Over 1100 fossil pollen time series from all continents

(except Antarctica)

Adapted from Mottl & Flantua et al. ([2021](#)) *Science*



Global rates of vegetation change

Adapted from Mottl & Flantua et al. ([2021](#)) *Science*



Global rates of vegetation change

Adapted from Mottl & Flantua et al. (2021) *Science*



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Global rates of vegetation change

Adapted from Mottl & Flantua et al. (2021) *Science*





DOI 10.1126.science.abg1685

Mottl & Flantua et al. (2021) *Science*



Continental Patterns

We do not have to stop at single
ecosystem property (e.g., RoC)

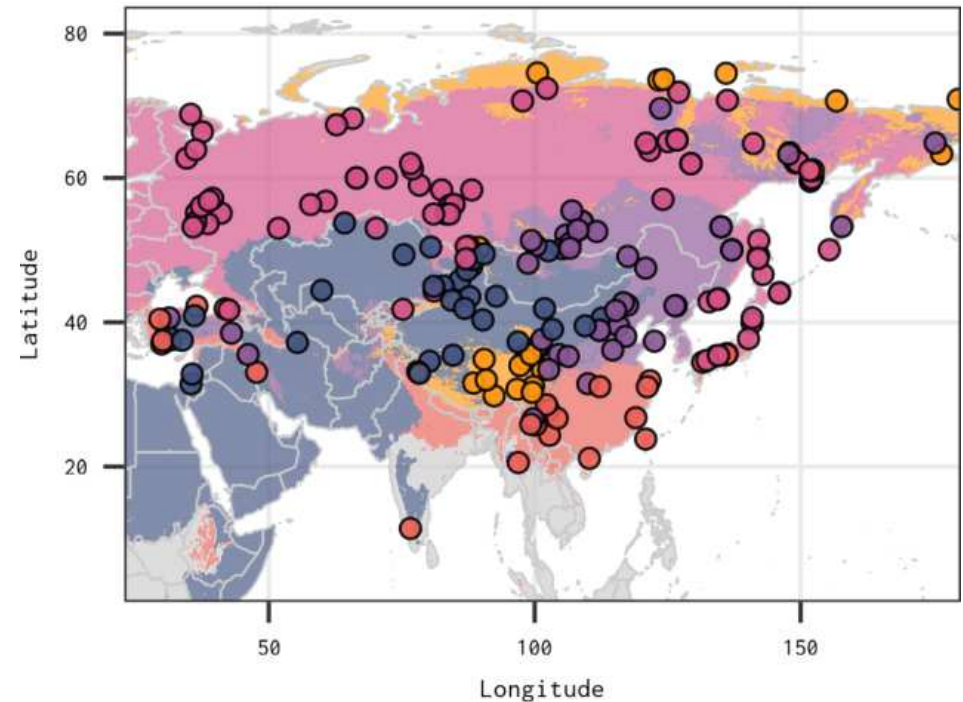


Continental Patterns

We do not have to stop at single ecosystem property (e.g., RoC)

Asia as a case study spanning diverse climates and vegetation types

Asia Case Study



Adapted from Bhatta et al. ([2023](#)) *Frontiers in Ecology and Evolution*



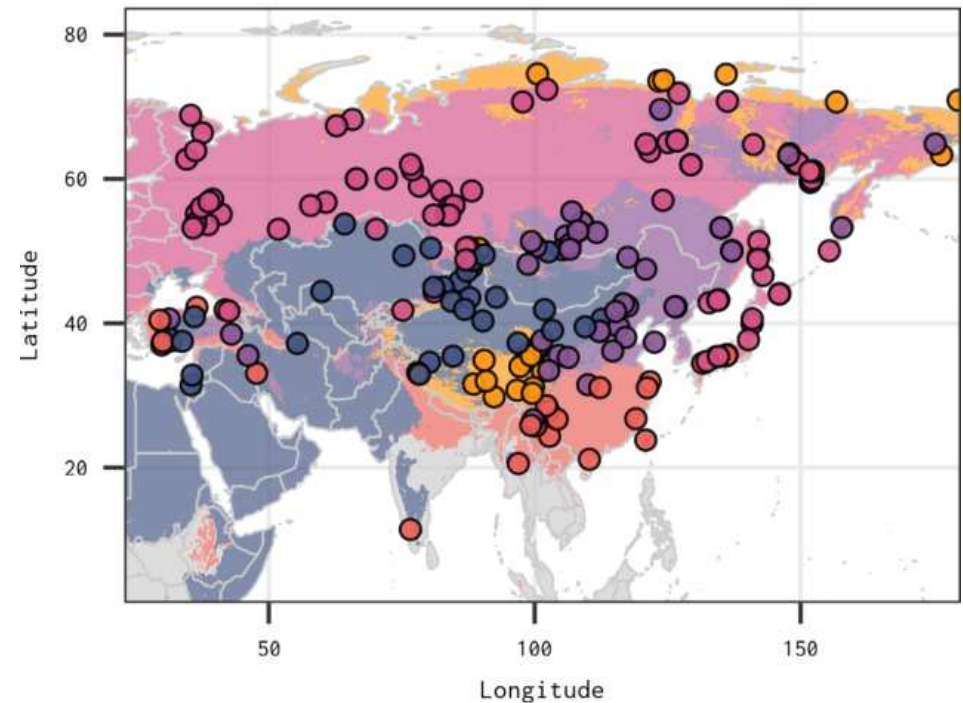
Continental Patterns

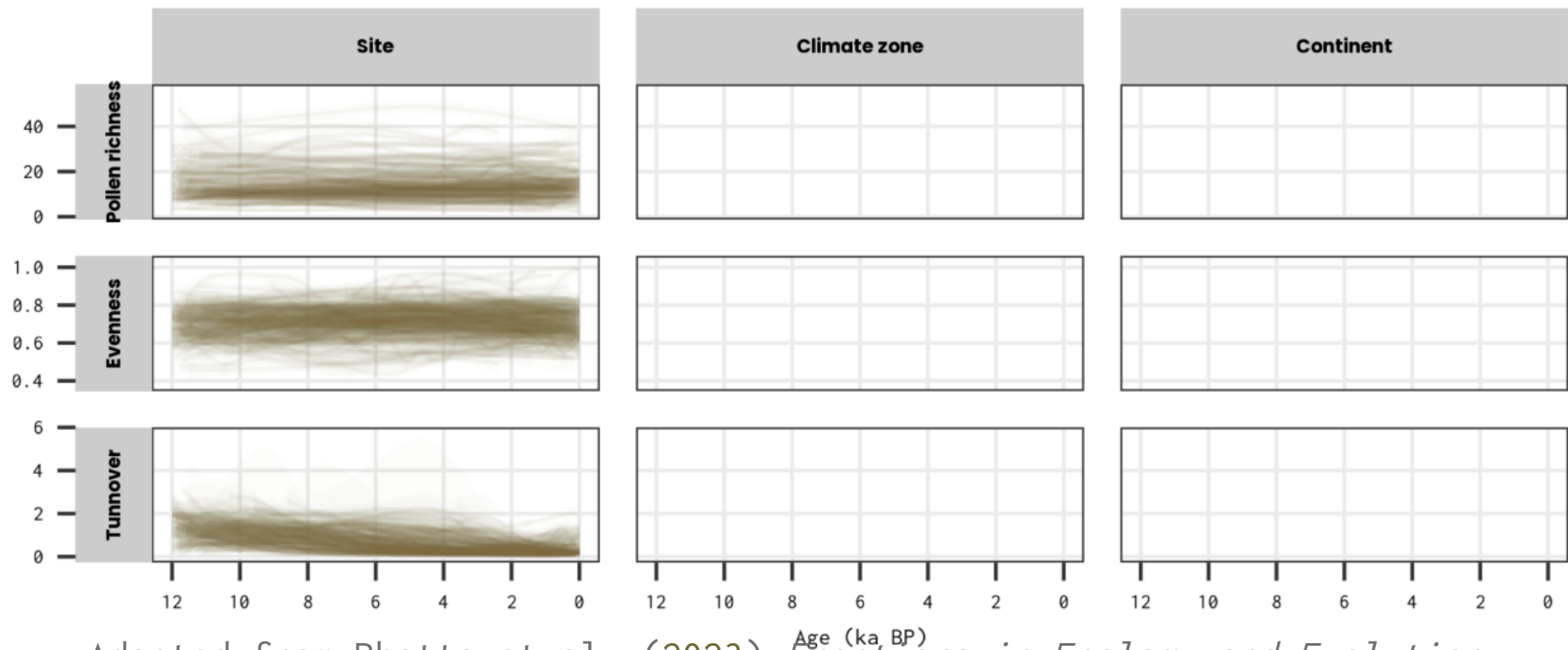
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Asia as a case study spanning diverse climates and vegetation types

Can we identify consistent continental trend of vegetation change across Asia during the Holocene?

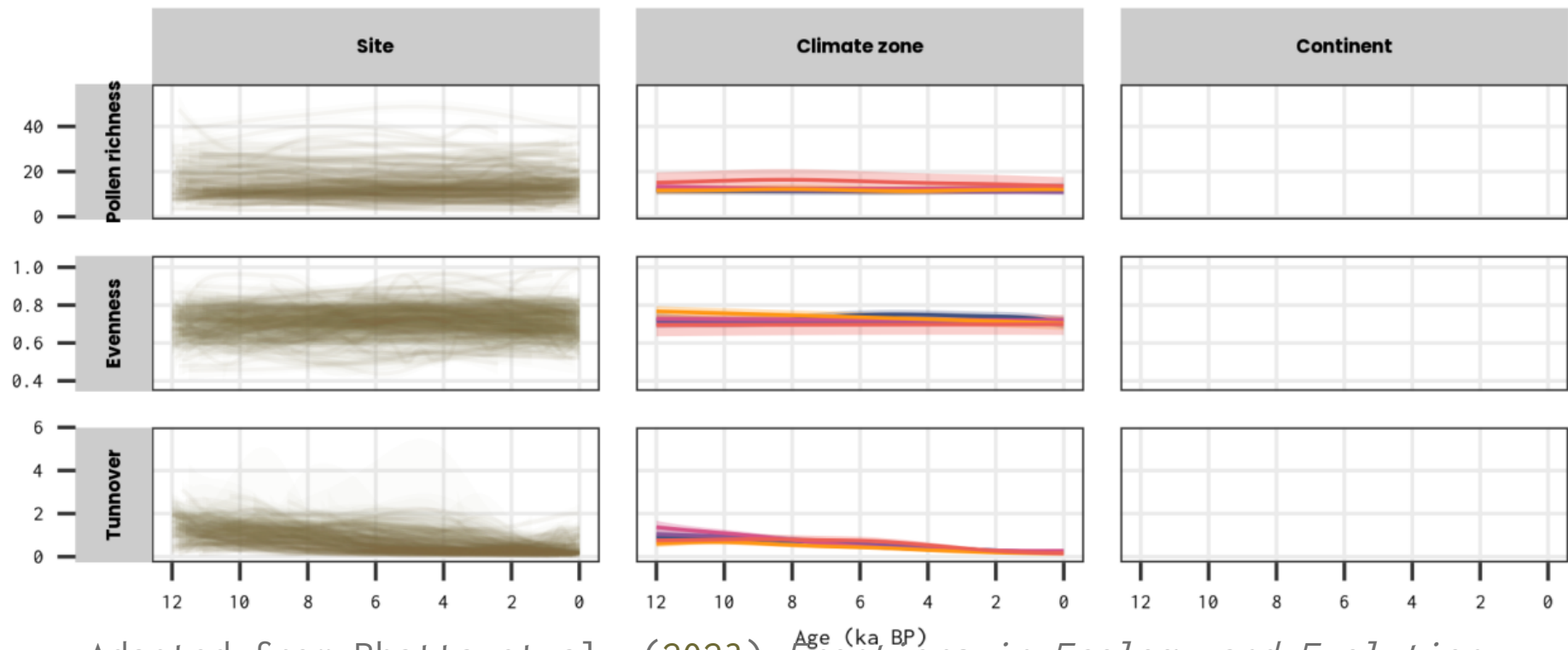
Asia Case Study





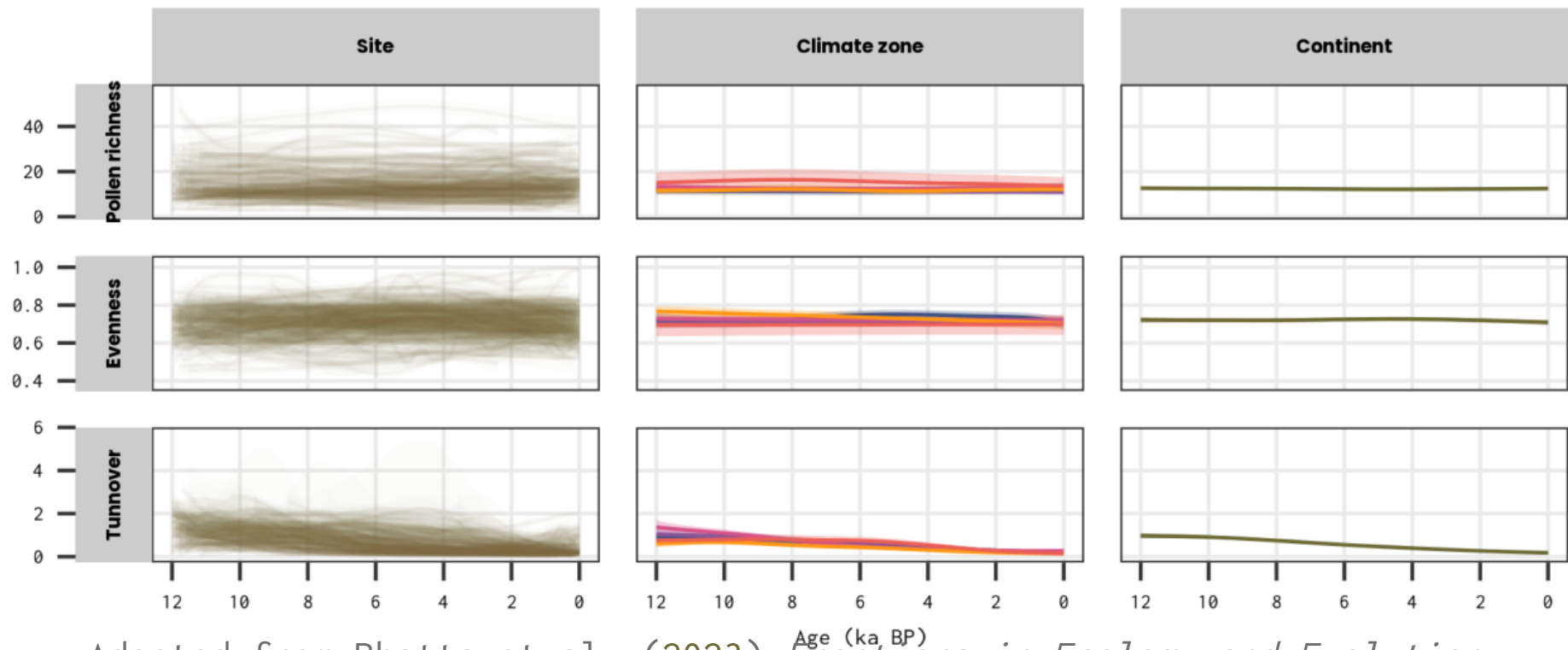
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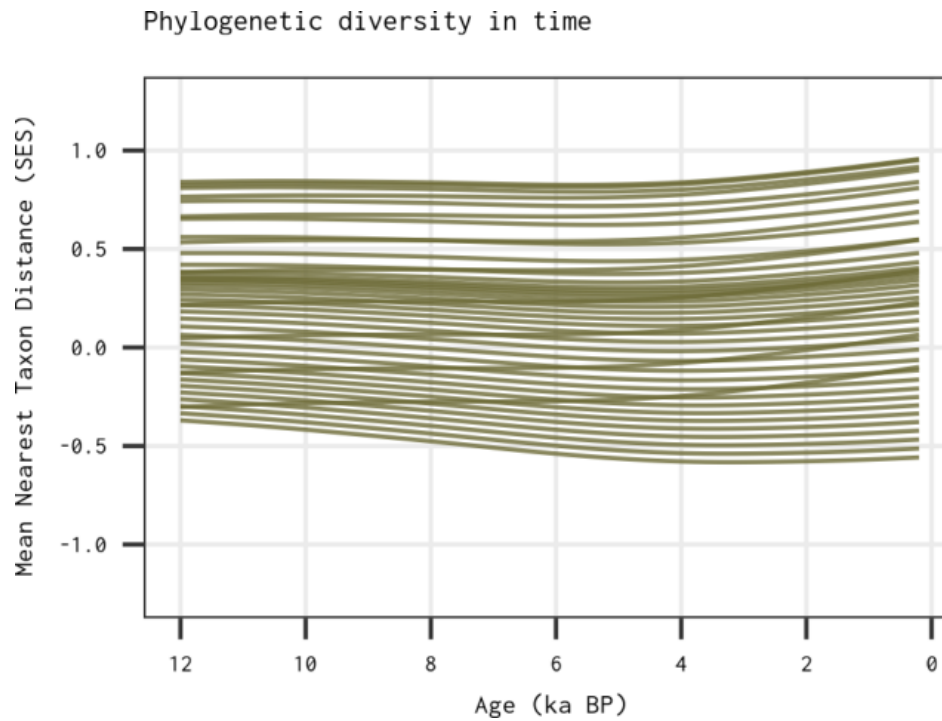
Adapted from Bhatta et al. ([2023](#)) *Frontiers in Ecology and Evolution*





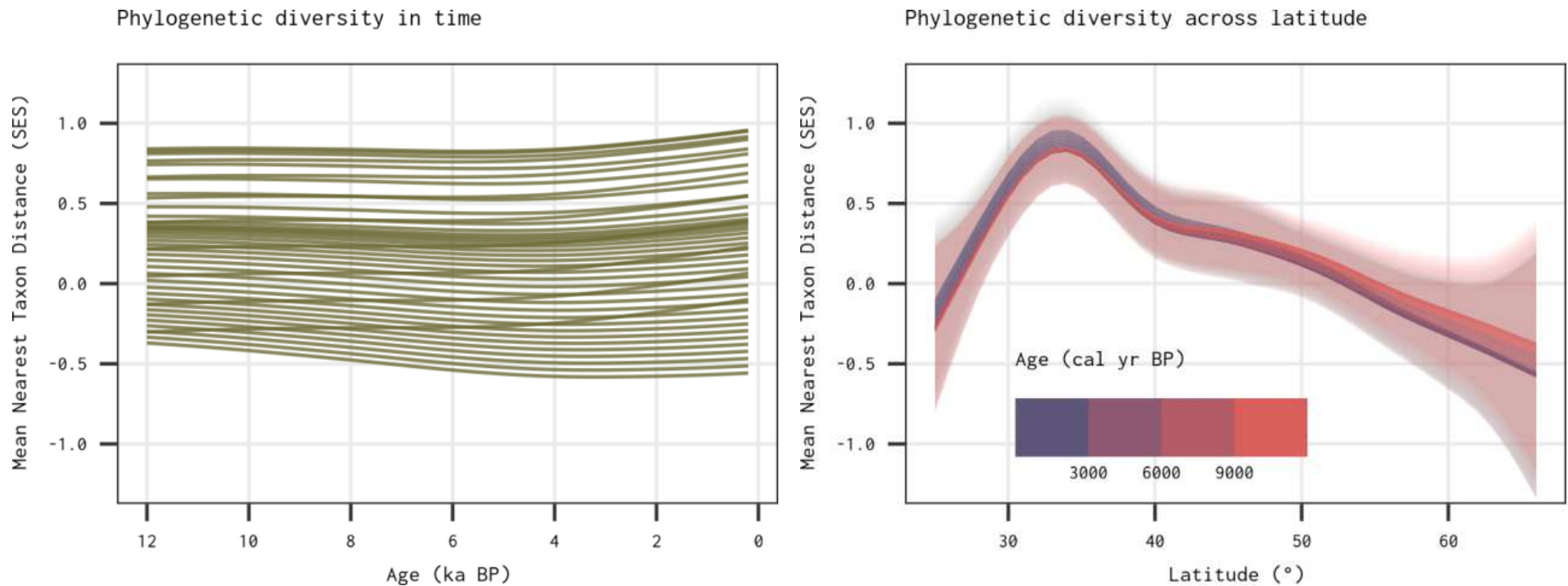
Adapted from Bhatta et al. ([2023](#)) *Frontiers in Ecology and Evolution*





Adapted from Bhatta et al. ([2024](#)) *scientific reports*





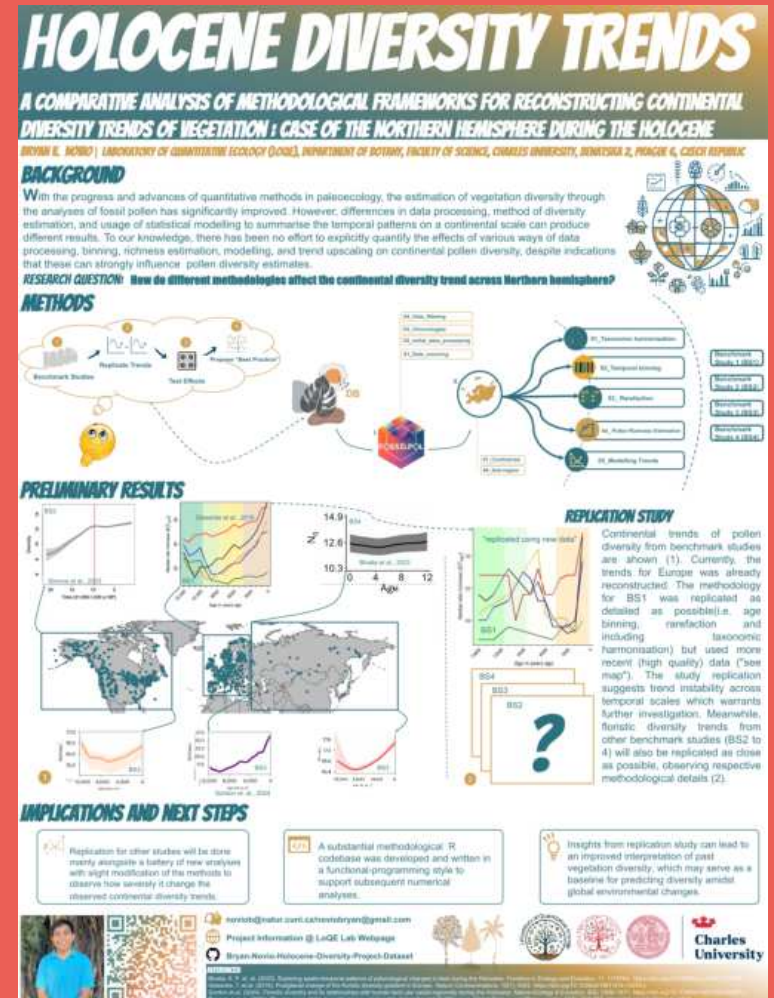
Adapted from Bhatta et al. ([2024](#)) *scientific reports*



Student intermezzo

Poster No. 2006

A Comparative Analysis of Methodological Frameworks for Reconstructing Continental Diversity Trends of Vegetation



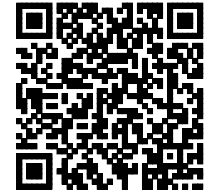


Functional Traits

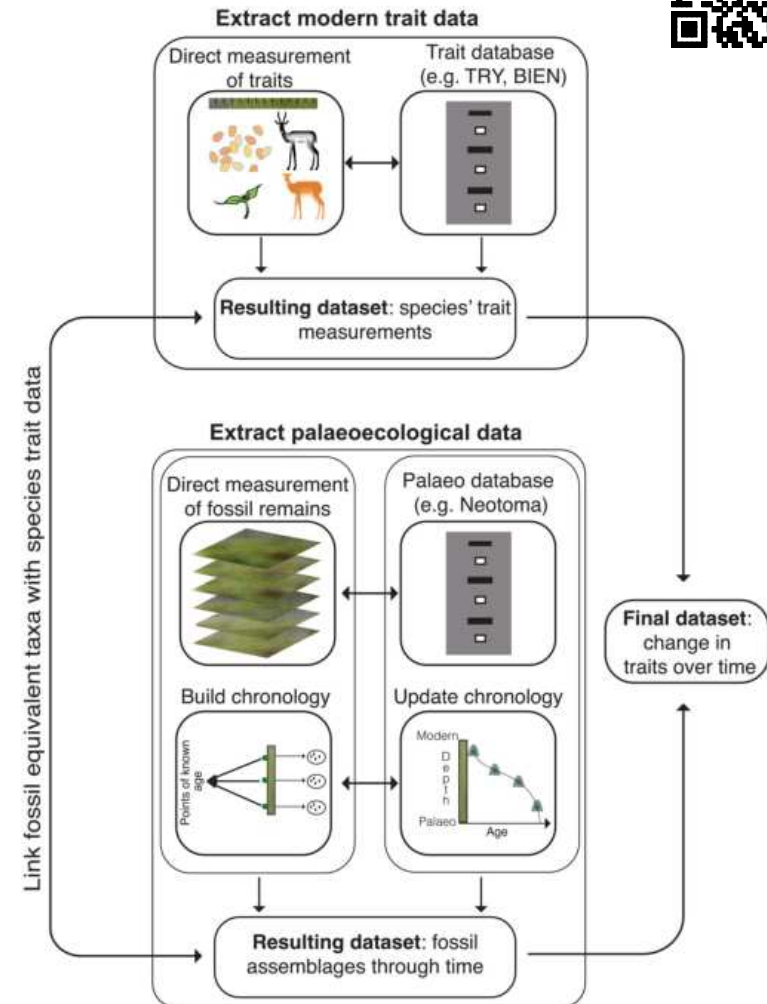
- Taxonomic identity $\neq$ ecological function
- Functional ecology allow to generalise across taxa, regions, and time
- Link between palaeoecological proxies and functional traits



Functional Traits



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- Functional ecology allow to generalise across taxa, regions, and time
- Link between palaeoecological proxies and functional traits



REVIEW | [Open Access](#) | [CC](#) [i](#)

Trait-based approaches as ecological time machines: Developing tools for reconstructing long-term variation in ecosystems

Kerry A. Brown [✉](#) M. Jane Bunting [✉](#) Fabio Carvalho [✉](#) Francesco de Bello [✉](#) Luke Mander [✉](#)
Katarzyna Marcisz [✉](#) Ondrej Mottl [✉](#) Triin Reitalu [✉](#) Jens-Christian Svenning [✉](#)

First published: 22 August 2023 | <https://doi.org/10.1111/1365-2435.14415> |

[VIEW METRICS](#)

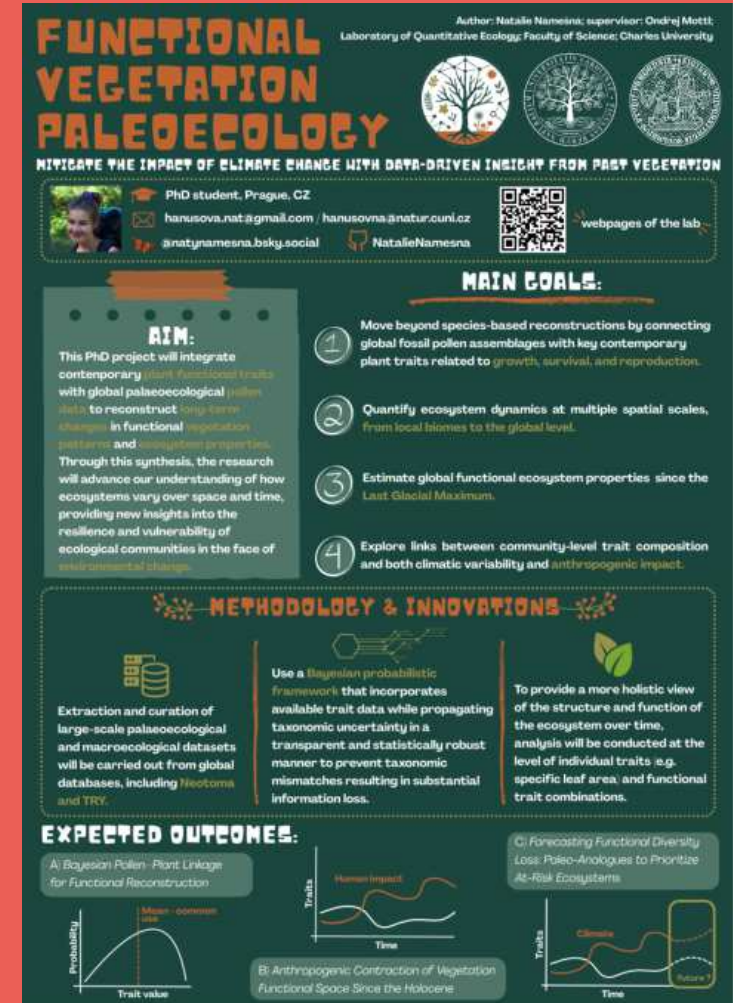
Figure 1 from Brown et al. (2023) *Functional Ecology*



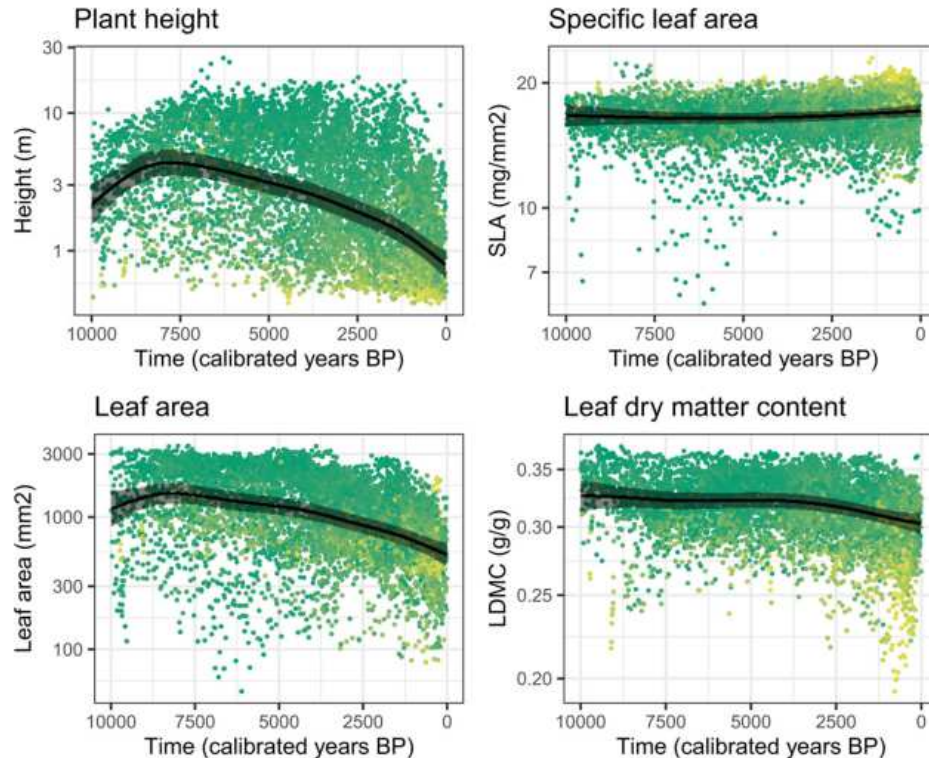
Student intermezzo

Poster No. 2005

Functional Vegetation Paleoecology: Mitigate the impact of climate change with data-driven insight from past vegetation



Traits in time



ECOLOGY LETTERS

LETTER | [Open Access](#) | [CC BY](#)

Pollen-based reconstruction reveals the impact of the onset of agriculture on plant functional trait composition

[Annegreet Veeken](#) ✉ [Maria J. Santos](#) [Suzanne McGowan](#) [Althea L. Davies](#) [Franziska Schrod](#)

First published: 11 July 2022 | <https://doi.org/10.1111/ele.14063> | [VIEW METRICS](#)

Editor: John Williams

Example for subcontinental scale **integration of functional traits** with **fossil pollen** data

Figure 4 from Veeken et al. (2022) *Ecology Letters*



Data Integration

Photo by [panumas nikhomkhai](#) from Pexels



Denmark example

*How has vegetation in **Denmark** remain **functionally stable** under **climate** change from the **LGM** to the **present***



Denmark example

*How has vegetation in **Denmark** remain **functionally stable** under **climate** change from the **LGM** to the **present***

Data Needs

- Fossil pollen and contemporary vegetation plots
- Functional traits (e.g. height, SLA, seed mass)
- Climate data (past and contemporary)



Denmark example

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Data Needs

- Fossil pollen and contemporary vegetation plots
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- Climate data (past and contemporary)

Obstacles

- ✗ Several data sources (formats, tools),
- ✗ Taxonomies mismatch across datasets
- ✗ Technically challenging (with limited reproducibility)



VegVault + {vaultkeeper}



*How has vegetation in **Denmark** remain **functionally stable** under **climate** change from the **LGM** to the present?*

With few lines of code, you can get the data you need to answer that!



```

1 data_denmark ←
2   vaultkeeper :open_vault(
3     path = "<path_to_VegVault>"
4   ) ▷
5   vaultkeeper :get_datasets() ▷
6   vaultkeeper :select_dataset_by_geo(
7     lat_lim = c(54.5, 58.0),
8     long_lim = c(8.0, 13.0)
9   ) ▷
10  vaultkeeper :get_samples() ▷
11  vaultkeeper :select_samples_by_age(
12    age_lim = c(0, 20000) # 20,000 BP
13  ) ▷
14  vaultkeeper :get_taxa(
15    classify_to = "genus"
16  ) ▷
17  vaultkeeper :get_abiotic_data() ▷
18  vaultkeeper :get_traits(
19    classify_to = "genus"
20  ) ▷
21  vaultkeeper :extract_data()

```



VegVault database



- A global SQLite database linking interdisciplinary plot-based vegetation data!
- Links to functional traits, climate variables, and soil properties
- Built for exploring biodiversity dynamics across time & space
- Open Source & Reproducible
-  Learn more: bit.ly/VegVault



VegVault database



~110 GB | 32 tables & 87 variables | 480,000+ Datasets | 13M+ Samples | 110,000+ Taxa | 11M+ Trait values | 8 Abiotic variables



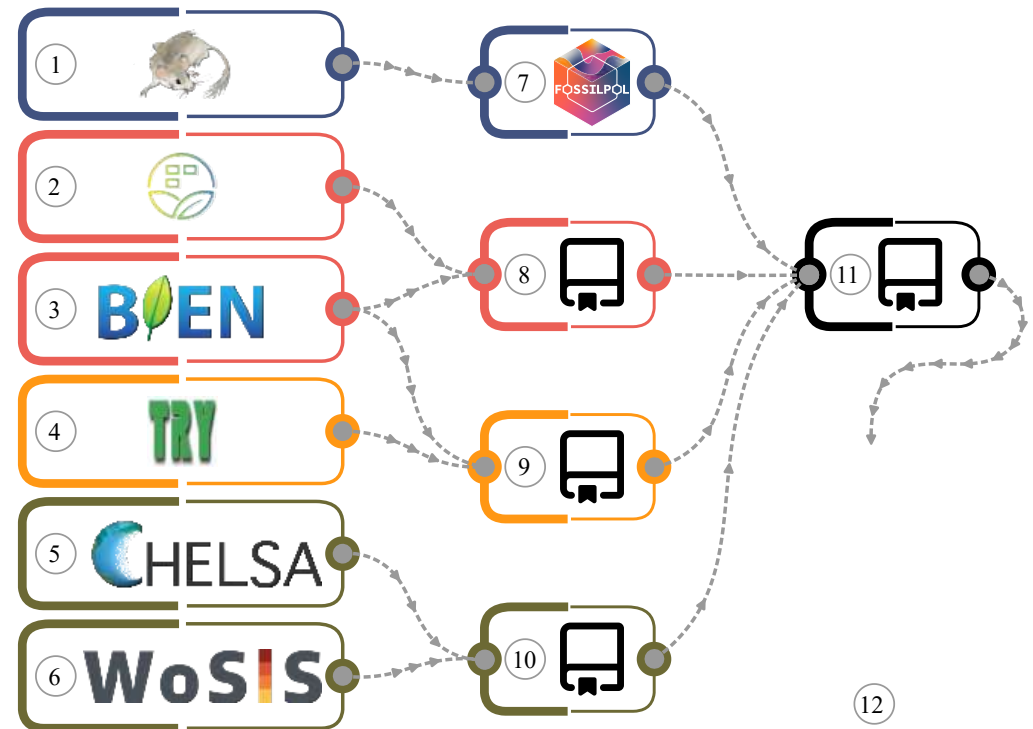
VegVault database



~110 GB | 32 tables & 87 variables | 480,000+ Datasets | 13M+ Samples | 110,000+ Taxa | 11M+ Trait values | 8 Abiotic variables

Data sources

-  **Paleoecological records** (*Neotoma*, *FOSSILPOL*)
-  **Contemporary vegetation** (*sPlotOpen*, *BIEN*)
-  **Functional traits** (*BIEN*, *TRY*)
-  **Climate & soil data** (*CHELSEA*, *WoSIS*)



{vaultkeepr} package



An Open Source R package 📦 - Your key 🔑 to **VegVault**



- ✓ No SQL required - just load & go!
- ✓ Filter by taxa, traits, time & space
- ✓ Tidyverse-compatible interface
- ✓ Directly usable in R for analysis
- 🔗 Learn more: bit.ly/vaultkeepr

R-CMD-check passing

 codecov 98%

CRAN not published



VegVault database



◆ Key Innovations

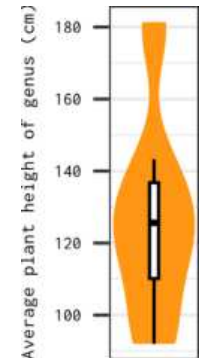
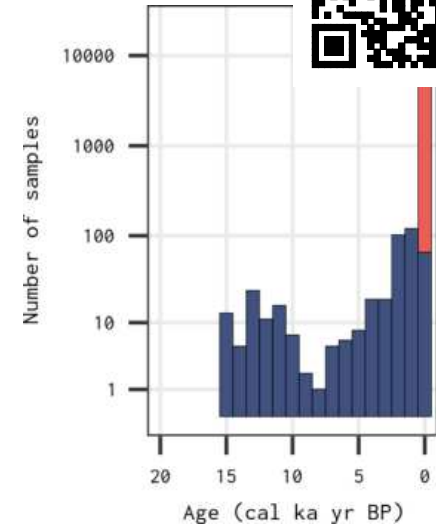
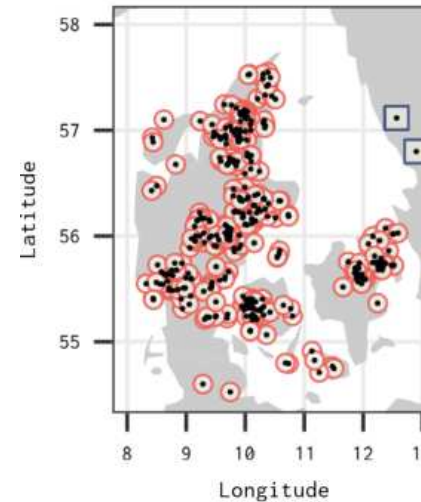
- Integration of interdisciplinary vegetation data into a single database
- Linking vegetation data with functional traits and abiotic variables
- User-friendly R package `{vaultkeeper}` for data access and extraction
- Open Source & Reproducible workflows for vegetation research



VegVault database

Key Innovations

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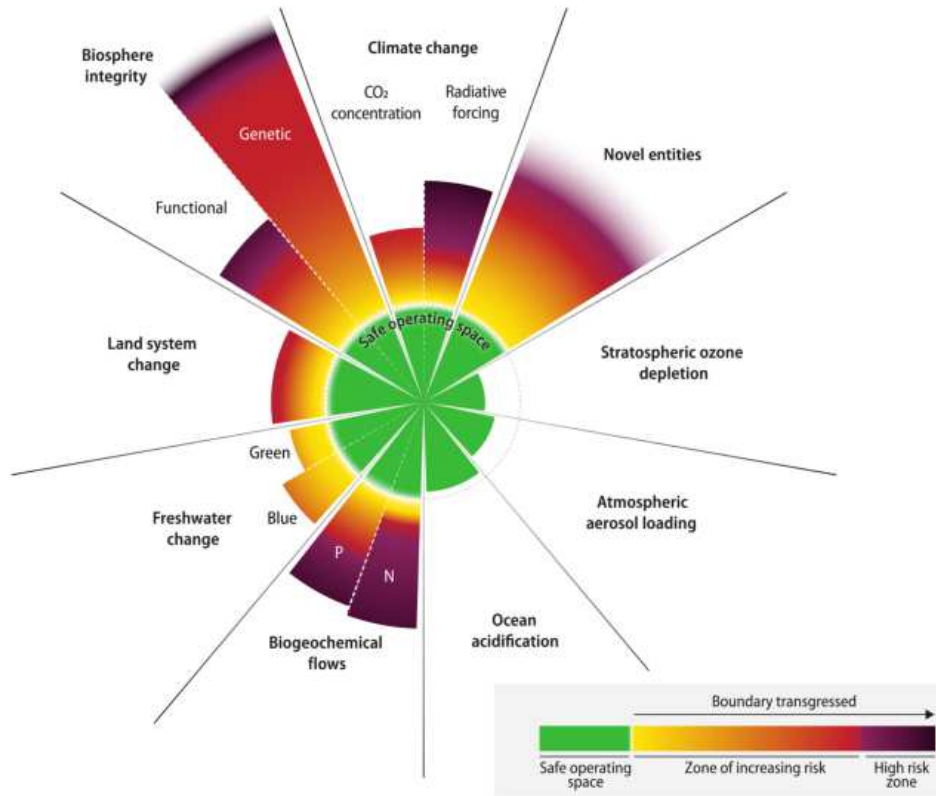
Planetary Boundaries



Photo by [Markus Spiske](#) from Pexels



Planetary Boundaries



Richardson et al. (2023) *Science Advances*



From Paleo to Policy



- Paleoecology offers long-term baselines for historical ranges of variability
- RAD (resist/accept/direct) framework for integrating paleoecological data into policy and decision-making
- Even if the past is an imperfect analogue, it is still essential for bounding plausible variability and diagnosing novelty.



REVIEW | [Open Access](#) |

Exploring the Interface Between Planetary Boundaries and Palaeoecology

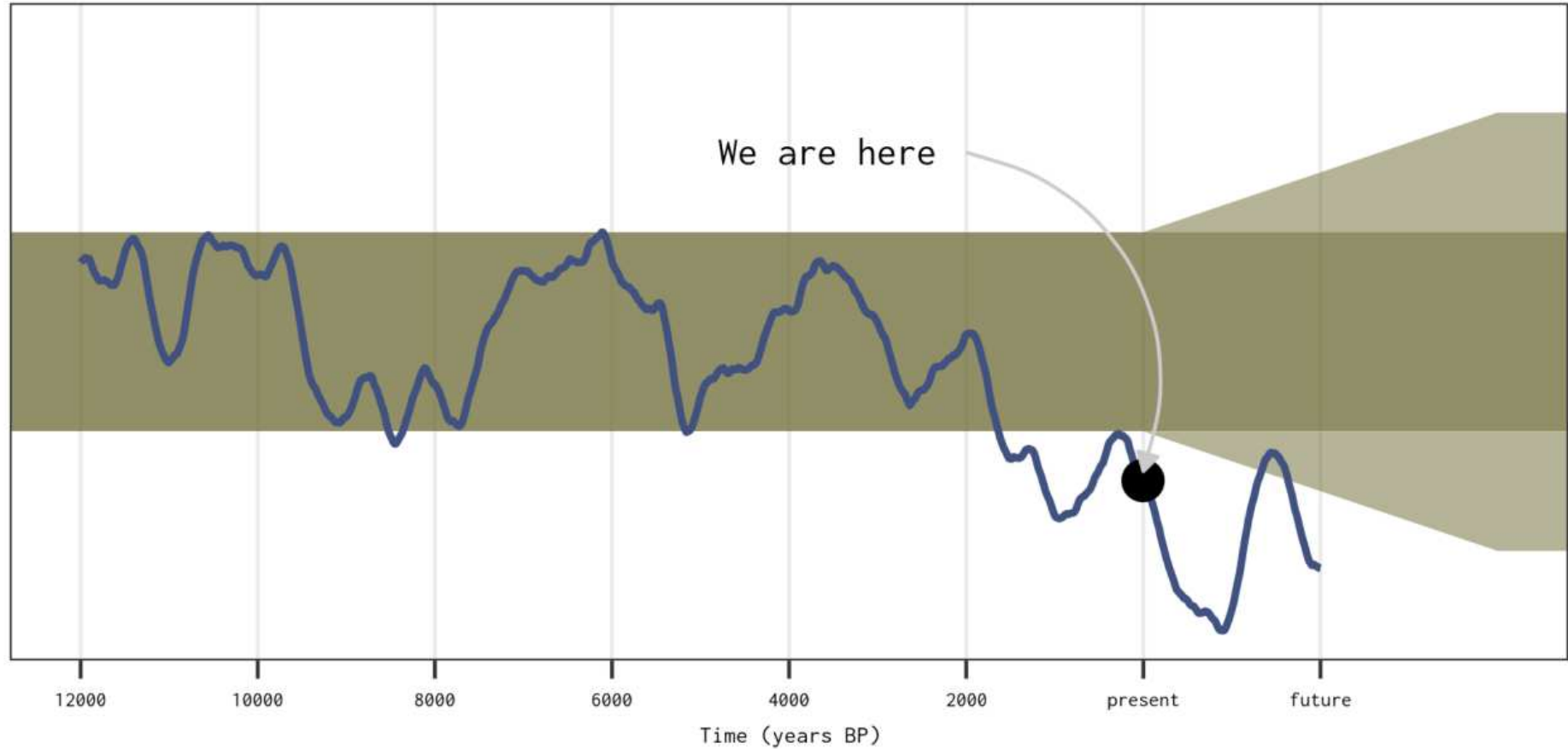
[This article relates to:](#)

[Lindsey Gillson](#) [Alistair Seddon](#), [Ondřej Mottl](#), [Ke Zhang](#), [Kelly Kirsten](#), [Peter Gell](#), [Rob A. Marchant](#), [Christoph Schwörer](#), [Estelle Razanatsoa](#), [Paul J. Lane](#), [Colin J. Courtney-Mustaphi](#), [John Dearing](#) ... [See fewer authors](#) ^

First published: 16 January 2025 | <https://doi.org/10.1111/gcb.70017> | [VIEW METRICS](#)



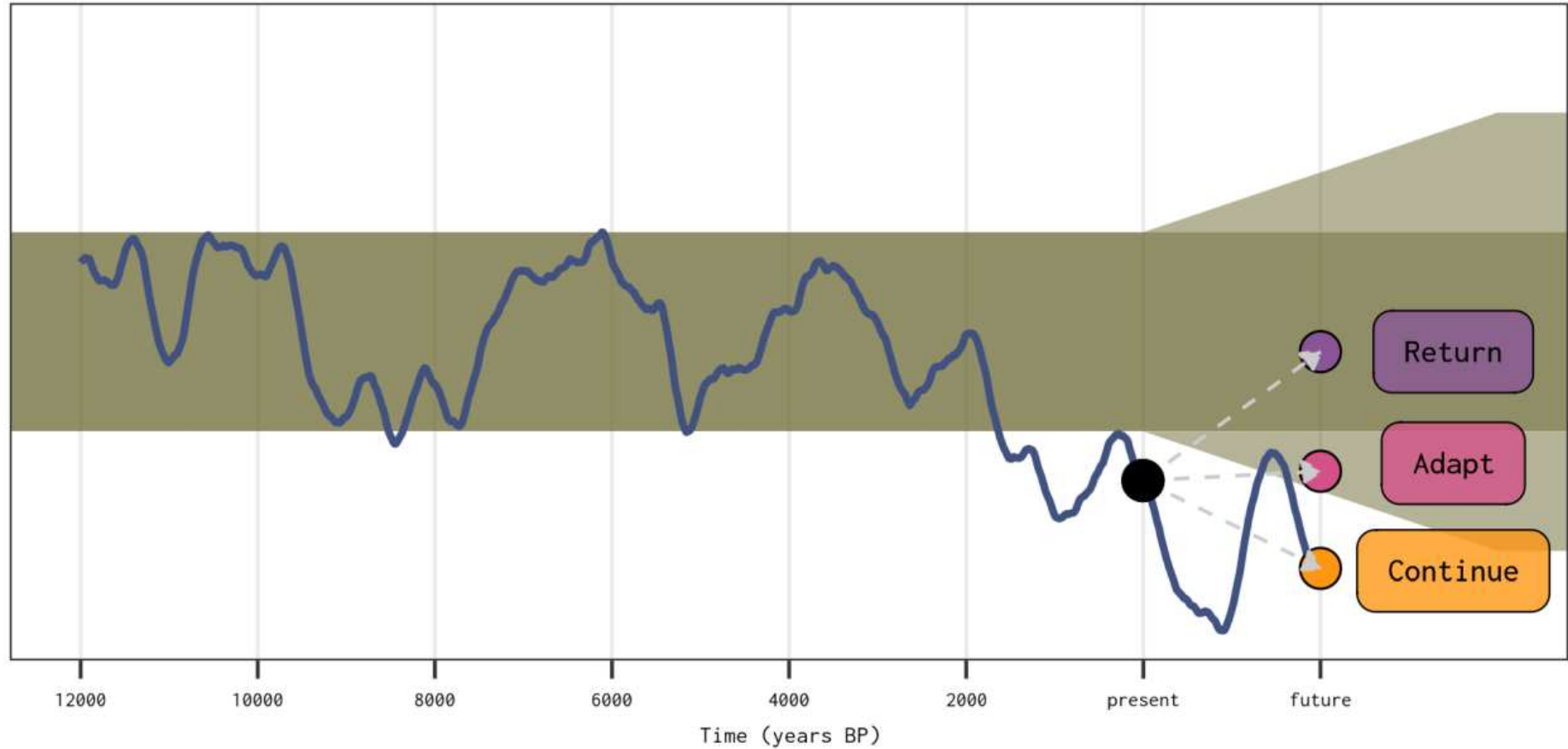
From Paleo to Policy



Gillson et al. ([2025](#)) *Global Change Biology*



From Paleo to Policy



Gillson et al. ([2025](#)) *Global Change Biology*



The time is ripe ... 👍



Photo by [Felix Mittermeier](#) from Pexels



The time is ripe ... 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales



Photo by [Felix Mittermeier](#) from Pexels



The time is ripe ... 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access



Photo by [Felix Mittermeier](#) from Pexels



The time is ripe ... 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access
- Ways to study various facets of biodiversity (taxonomic, functional, phylogenetic)



Photo by [Felix Mittermeier](#) from Pexels



The time is ripe ... 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access
- Ways to study various facets of biodiversity (taxonomic, functional, phylogenetic)
- Statistical methods to handle complex hierarchical data structures



The time is ripe ... 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access
- Ways to study various facets of biodiversity (taxonomic, functional, phylogenetic)
- Statistical methods to handle complex hierarchical data structures
- Computational power and Open Source tools for uncertainty-incorporating analysis



Photo by [Felix Mittermeier](#) from Pexels



***Paleoecological data is ready to inform
biodiversity-conservation and rewilding
actions!***



Photo by [Tudor Sorin](#) from Pexels



This presentation

is publically available!



Code



Slides



last slide

